COMATOSE CHILD

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What is coma?

- A state of unresponsiveness and unconsciousness
- Structural injuries to diffuse systemic abnormalities
- Optimal care requires timely intervention
- MENTAL STATE = Arousal + content

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Anatomy of mental status

- Ascending reticular activating system (ARAS)
  - Activating systems of upper brainstem, hypothalamus, thalamus
  - Determines the level of arousal
- Cerebral hemispheres and interaction between functional areas in cerebral hemispheres
  - Determines the intellectual and emotional functioning
- Interaction between cerebral hemispheres and activating systems

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Content of consciousness

- Sum of patient's intellectual (cognitive) functions and emotions (affect)
  Sensations, emotions, memories, images, ideas (SEMII)
- Depends upon the activities of the cerebral cortex, the thalamus & their interrelationship
- Lesions of these structures will diminish the content of consciousness (without changing the state of consciousness)

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Pathophysiology

- The ascending reticular activating system controls ones level of alertness or arousal
- ARAS is comprised of the rostral brainstem tegmentum, diencephalon and associated cortical projections
- Sleep centre is the preoptic area of the hypothalamus
- Damage to the ARAS and associated structures or both structures is usually necessary to cause a comatose state.
- Left hemisphere dysfunction per se can lead to coma

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Pathophysiology

- Brain tolerates only limited physical/metabolic injury
- Longer the failure, greater the risk of permanent neurologic impairment
- Depth of coma assessed by level of consciousness
- Arousal: degree to which an individual can interact with the environment
- Awareness: depth and content of arousal state

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Definitions of levels of consciousness

- Clouding of consciousness – impaired capacity to think clearly
- Delirium – disturbed consciousness with motor restlessness, disorientation and hallucination
- Obtundation – reduced alertness, appears to sleep but responds to verbal and/or tactile stimuli
- Stupor – markedly reduced alertness, only responds to noxious stimuli
- Coma – no response to even noxious stimuli and will not utter understandable words

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Etiologies…

- **1. Metabolic – Toxic**
  - Hypoxemia – ischemia
  - Metabolic disorders
  - Fluid and electrolyte imbalances
  - Infection
  - Exogenous toxins and poisons
  - Paroxysmal disorders

2. Structural – Intrinsic
  - Trauma
  - Neoplasms
  - Vascular disease
  - Hydrocephalus

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Identification of cause

Coma may present as part of:

- progression of a known neurological illness
- unpredictable consequence
- complication of a known systemic disease
- result from a totally unexpected event or illness...

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Initial assessment and resuscitation…

- Evaluation of airway patency and ventilation
- Status of patient's perfusion
- Core temperature
- GCS score
- Respiratory pattern
- Hemodynamics
- Specific examination

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Raised ICP

- **Monro-Kellie doctrine** – A small change in the volume of any compartment must be accompanied by an equal and opposite change in another compartment or ICP must rise.
- Raised ICP leads to regional or global ischemia
- Complete obliteration of arterial blood flow leads to brain death
- Newborn child: ICP is 4.5 - 5 cm, adults 15-20 cms.
- Maintenance of CPP important in prevention of ischemia
- CPP = MAP - ICP

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Management

- Treatment of underlying cause
- Cerebroprotective measures
- Medications
- Monitoring during therapy

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Key points

- Early control of airway and breathing mandatory
- Meticulous maintenance of normotension, normovolemia, normal temperature and metabolic parameters essential
- Control seizures and agitation
- Overzealous hyperventilation can lead to neurological damage

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GLASGOW COMA SCALE

Behaviour	Response	Score
Eye opening response	Spontaneously	4
	To speech	3
	To pain	2
	No response	1
Best verbal response	Oriented to time, place, person	5
	Confused	4
	Inappropriate words	3
	Incomprehensible sounds	2
	No response	1
Best motor response	Obeys commands	6
	Localizes pain	5
	Flexion withdrawal from pain	4
	Abnormal flexion	3
	Abnormal extension	2
	No response	1
Total Score		15

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BLANTYRE COMA SCALE

BEST MOTOR RESPONSE	
Localizes painful stimulus	2
Withdraws limb from pain	1
Non-specific or absent response	0
VERBAL RESPONSE	
Appropriate cry	2
Moan or inappropriate cry	1
None	0
EYE MOVEMENT	
Directed (eg. Follows mother's face)	1
Non directed	0

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